

MIT (An Autonomous Institute Affiliated to SPPU) Alandi (D), Pune – 412105 Department of Computer Engineering (Software Engineering)	Academy of Engineering	CA Activity-03	
		ACADEMIC YEAR	2025-26
		SEM / TRI	II (Regular)
		CLASS & DIVISION / BLOCK	First Year Div All

ACTIVITY NAME	Design/Ideation Assignment on UNO Goals		
COURSE	CREATIVE TECHNOLOGIES (2301183)	Lab Activity NO.	03 (Three)
COURSE INSTRUCTORS	VINAYAK KULKARNI & TEAM	DATE	25-01-2026

STUDENT NAME	Uday Madankar	PRN	202501030039
TRACK	PR(14)	TRACK TEACHER	M rs. Priyanka Nalawade Mam

Que. No.	Question Description	Marks	CO No.	BT Level
Q1	We are celebrating Year 2026 as the International Year of Women Farmers as declared by United Nations. You have to go through Question statement provided on next page. Develop an idea or propose a solution by applying your track technology; Submit a solution in form of a Poster or Powerpoint presentation or Web page. You must follow exact submission guidelines provided in problem statement	05	CO1 CO3	L3
Last date of Submission: 15 April 2025/Hard Deadline				

RUBRICS

No	Criteria	Excellent (Marks)	Satisfactory (Marks)	Average (Marks)
1.				
2.	Problem understanding (01)	Correct (01)	Complete(0.5)	Not Correct(00)
3.	Problem Solution (02)	Better (02)	Good(01/005)	Not Correct(00)
4.	Originality (01)	Original (01)	Partial (0.5)	Not Correct(00)
	Presentation (01)	Optimized (01)	Good (0.5)	Not Correct(00)

03

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Instructions:

1. Read this assignment page carefully
2. Check references given at the end
3. Attach this statement to your submission
4. Take Hardcopy of this report, get sign & Grade (10) from Track Teacher
5. Submission on Moodle of respective Track. (Pdf Format).
6. All FY students appearing for course Creative Technologies must do this submission individually.
7. Less originality/Copying contents/pure Chat GPT contents will reduce marks to lowest level.

(Vinayak B Kulkarni)

CourseInstructor

Sign with Date

INSTRUCTIONS & FORMAT OF SOLUTION	
1.	- Read TheAssignment pagecarefully, (next page, Track wise problems) - Read references and go through You tube videos given at the end This is internationalassignment, beingsolvedsimilar problemsacross the world. Some of Best assignments will be shared in Public as Indian Initiative
2.	- Select problem domain as per your division, which is given on next pages - Solve problem of your track only; otherwise, it will not be graded - Select your Problem statement individually - Search the content related to Problems and select problem area, propose solution
3.	- Develop an original theoretical solution for problem selected from your assigned domain(Division wise), it can be in the form of Rough Idea, Written expression of Thought, Diagrammatic representation, Sketch etc. - Present problem solution by applying this course contents learnt by you. - Your Solution must be in line with Sustainable Development Goal given in your Division Problem topic [Strictly Solve Problem Allocated to your Track] - General content/Very basic information /Content Without Problem Statement and Identified Solution will be graded to low marks Mandatory contents: - Roll No., PRN, Name, Track - Search the contents and find Problem Area - Title for Problem Solution - Introduction - Need for Solution (How problem is Important) - Proposed solution - Possible Effectiveness of solution/Benefits - Conclusion - References
4.	Submission- - Itis individual submission - Weightage CA marks – 05/35 - Select Submission format (Poster/PPT/Web page in pdf format) and present in decent manner
5.	Submission Format on Moodle – pdf [Any One] after grading
	i) Power point presentation (PPT) - 10 Slides Max- Pdf submission ii) Poster: A4 or A5 page size – pdf submission iii) Web page- standard size- Pdf submission and Html + Files in folder
6.	Submission Deadline:
	15 April2026is hard deadline, - No submissions will be accepted after this date marking zero marks.
7.	Don'ts: Less originality/Direct Copying contents/Pure Chat GPT contents will reduce marks to lowest level.
Assignment Statement on next page (Track Wise)	

ASSIGNMENT STATEMENT



INTERNATIONAL YEAR OF THE
WOMAN FARMER
2026

Introduction:

International Year of Women Farmers

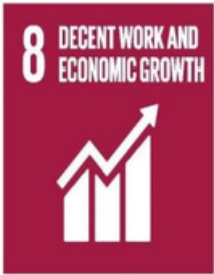
Women play essential roles across agrifood systems, producing, processing and trading food that sustains families, communities, and economies. Yet their contributions often go unrecognized, and persistent inequalities limit their access to resources, opportunities and decision making.

The International Year of the Woman Farmer (IYWF 2026) highlights these realities, calls for collective action and increased investments to close gender gaps, strengthen women's livelihoods and promote their leadership across agrifood value chains. By advancing gender equality and creating conditions for women and girls engaged along agrifood value chains to thrive, the Year contributes to building fairer, more inclusive and sustainable agrifood systems for all.

Empowered women farmers are key agents of change in transforming agrifood systems. Closing gender gaps in access to land, finance, technology, education and decision-making enhances the well-being of women and their households, boosts productivity, strengthens resilience to climate change and drives progress towards the Sustainable Development Goals (SDGs).

Women farmers are all women working in agrifood systems in different capacities across all segments of value chains. This includes farmers, producers, peasants, family and smallholder farmers, seasonal labourers, fishers, fish workers, beekeepers, pastoralists, foresters, processors, traders, traditional knowledge holders, women agricultural sciences, formal and informal workers, and entrepreneurs. They encompass women in all their diversity, including young and older women, Indigenous women and women in local communities, women with disabilities, refugee and displaced women, and others.

The definition is inclusive of women in both formal and informal work, recognizing their contributions regardless of land ownership or employment status. It reflects the diverse and essential roles all women play in sustaining agrifood systems including through leadership, care and domestic work, while supporting food security, contributing to economic prosperity, and improving nutrition and livelihoods for their families and communities.

Track	SDG	Assignment Question (Develop an application idea or propose a solution using)
AI [Artificial Intelligence]		- Understand what is UNO Sustainable Development Goal Zero Hunger - Understand Problems of Women farmers related to Zero Hunger problem - Select any one problem - Apply AI technology and generate an idea/solution - Develop a ppt/poster for your solution
PR [Prompt Engineering]		- Understand what is UNO Sustainable Development Goal Good Health & Well Being - Understand Problems of Women farmers related to Goal Good Health & Well Being - Select any one problem - Apply PR engineering technology and generate an idea/solution - Develop a ppt/poster for your solution
AR [Augmented Reality]		- Understand what is UNO Sustainable Development Goal Climate & Action - Understand Problems of Women farmers related to Goal Climate & Action - Select any one problem - Apply AR technology and generate an idea/solution - Develop a ppt/poster for your solution
BCT [BlockChain Technology]		- Understand what is UNO Sustainable Development Goal Decent Work and Economic Growth - Understand Problems of Women farmers related to Goal Decent Work and Economic Growth - Select any one problem - Apply BCT technology and generate an idea/solution - Develop a ppt/poster for your solution
RBX [Robotics] [Aggro Industry]		- Understand what is UNO Sustainable Development Goal Industry Innovation and Infrastructure - Understand Problems of Women farmers related to Goal Industry Innovation and Infrastructure - Understand Challenges for Agricultural Industry - Select any one problem - Apply Robotics technology and generate an idea/solution - Develop a ppt/poster for your solution
CC [Cloud Computing] [Food Industry]		- Understand what is UNO Sustainable Development Goal Industry Innovation and Infrastructure - Understand Problems of Women farmers related to Goal Industry Innovation and Infrastructure - Understand Challenges for Food Industry - Select any one problem - Apply Cloud Computing technology and generate an idea/solution - Develop a ppt/poster for your solution

Track	SDG	Assignment Question (Develop an application idea or propose a solution using)
IOT [Internet of Things] [Dairy Industry]		- Understand what is UNO Sustainable Development Goal Industry Innovation and Infrastructure - Understand Problems of Women farmers related to Goal Industry Innovation and Infrastructure - Understand Challenges for Dairy Industry - Select any one problem - Apply IOT technology and generate an idea/solution - Develop a ppt/poster for your solution
DRN [Drone Technology] [Aggro Industry Automation]		Understand what is UNO Sustainable Development Goal Industry Innovation and Infrastructure Understand Problems of Women farmers related to Goal Industry Innovation and Infrastructure Understand Challenges for Agricultural Industry Automation using Drones - Select any one problem - Apply Drone technology and generate an idea/solution - Develop a ppt/poster for your solution
EV [Electric Vehicles] [Aggro Industry Automation]		Understand what is UNO Sustainable Development Goal Industry Innovation and Infrastructure Understand Problems of Women farmers related to Goal Industry Innovation and Infrastructure Understand Challenges for EV for Agricultural Industry Automation - Select any one problem - Apply EV technology and generate an idea/solution - Develop a ppt/poster for your solution
AECE [Affordable & Clean Energy]		- Understand what is UNO Sustainable Development Goal Affordable & Clean Energy - Understand Problems of Women farmers related to Goal Affordable & Clean Energy - Understand Challenges for Energy in Agricultural Industry - Select any one problem - Apply AECE technology and generate an idea/solution - Develop a ppt/poster for your solution
ZEB [Zero Energy Building] [Storage facilities & Ware houses]		- Understand what is UNO Sustainable Development Goal Sustainable Cities and Communities - Understand Problems of Women farmers related to Goal Sustainable Cities and Communities - Understand Challenges for constructing Storage Facilities & Ware houses - Select any one problem - Apply ZEB technology and generate an idea/solution - Develop a ppt/poster for your solution

References: [Must Read before Solution]

1. UNO Goals or Sustainable Development Goals

<https://sdgs.un.org/goals>

<https://unric.org/en/united-nations-sustainable-development-goals/>

<https://youtu.be/4XVYucv2tuA?si=jARDxcIzAS82iKGh>

<https://youtu.be/hsomzuNWD8E>

<https://youtu.be/v-Afm04x2Ow>

2. International Year of Women Farmers-2026

<https://iywf2026.com/>

<https://www.global-agriculture.com/farming-agriculture/international-year-of-the-woman-farmer-2026-a-call-to-recognise-and-empower-women-in-agriculture/>

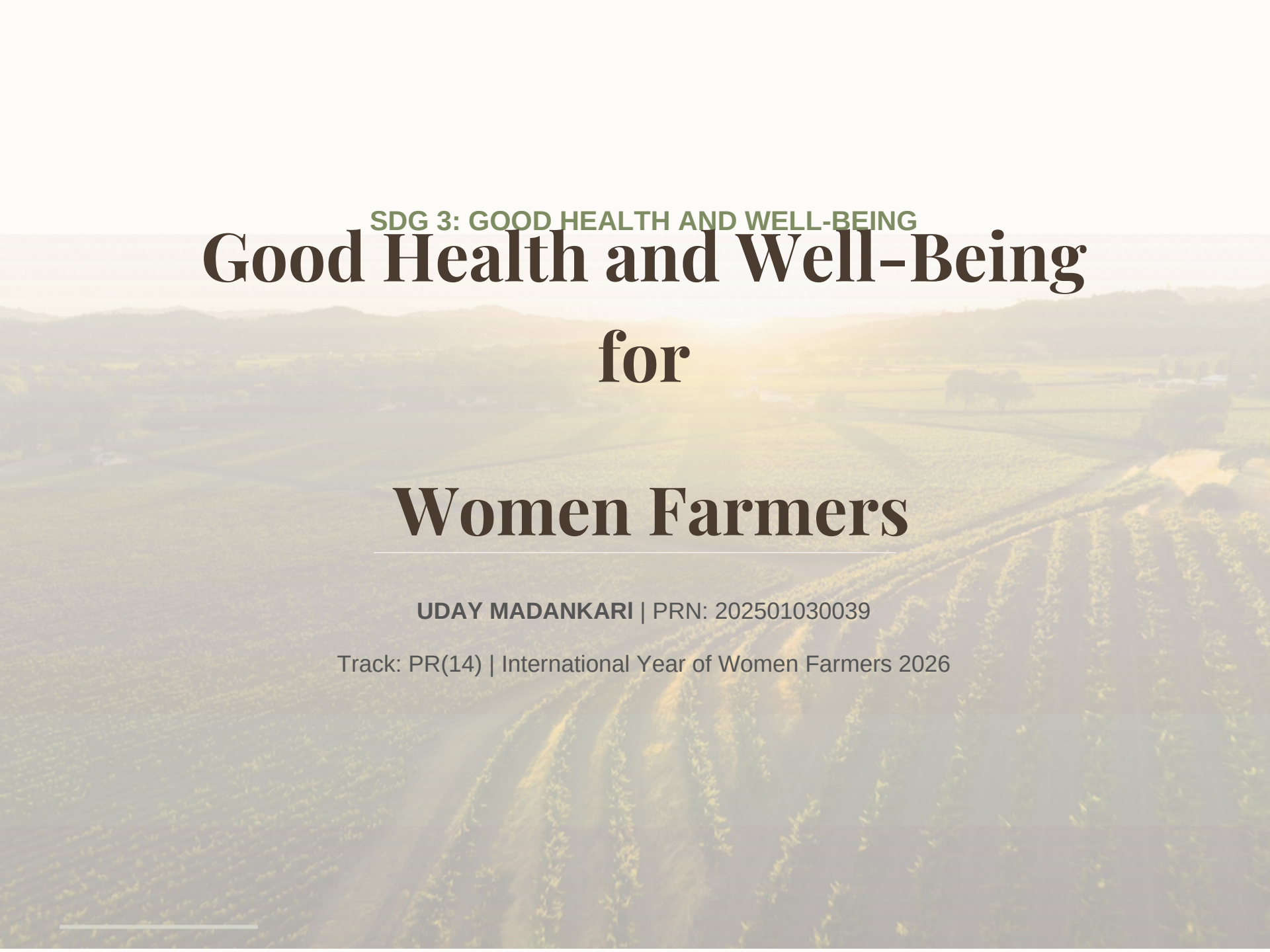
<https://www.farms.com/ag-industry-news/women-in-agriculture/breaking-barriers-why-2026-is-the-year-of-the-women-farmer-750.aspx>

<https://youtu.be/aNf1Boky1M4>

<https://youtu.be/Uc8-iyArlL0>

<https://youtu.be/7dyfOrWrcoA>

<https://youtu.be/BUj5Xh8nTN8>



SDG 3: GOOD HEALTH AND WELL-BEING

Good Health and Well-Being for

Women Farmers

UDAY MADANKARI | PRN: 202501030039

Track: PR(14) | International Year of Women Farmers 2026

Introduction: The Unsung Heroes

- 🌱 **Vital Economic Pillars:** Women farmers play a foundational role in global agriculture, rural development, and sustained economic growth.
 - 📦 **A Hidden Crisis:** Despite massive contributions, their occupational health and personal safety are frequently overlooked in global policy.
 - 📅 **UN Recognition:** The United Nations has declared 2026 as the **International Year of Women Farmers** to rectify this neglect.
 - 🛡️ **Our Focus:** Implementing smart engineering solutions to protect the well-being of these essential workers.
-

SDG 3 Objectives & Key Goals

Primary Objective

Ensure healthy lives and promote well-being for all ages. In the agricultural context, this means transforming farming from a hazardous chore into a safe, sustainable profession.

Target Outcomes

- Reduce occupational diseases and risks.
- Improve access to primary healthcare.
- Promote rigorous safety standards.
- Enhance mental and physical well-being.

Major Health Challenges Identified

- ⚗ Exposure to harmful pesticides and chemicals.
- 🦴 Chronic back pain and joint issues from manual labor.
- ⚙ Severe heat stress and chronic dehydration.
- 👤× Critical lack of protective equipment (PPE).
- 🏠 Limited awareness and access to facilities.



The Critical Need for Intervention



Safety First

Ensuring women farmers are shielded from immediate occupational hazards and long-term toxicity.



Productivity

A healthy workforce is an efficient one. Reducing injuries directly boosts national food security.



Gender Equality

Promoting equity by designing tools and systems specifically for the female physiological needs.

The Smart Health & Safety Kit

Our proposed engineering solution integrates lightweight ergonomics with modern IoT sensing to create a comprehensive safety shield for the woman farmer.

Ergonomic Tools: Designed for female anatomy.

Solar Smart Hat: Active cooling & shade.

Monitoring Band: Vital signs tracking.

Mobile App: Real-time alerts & guidance.



Technological Core: IoT & Solar

We leverage cutting-edge Industry 4.0 tech to protect the most traditional of sectors:



IoT: Wearable bands monitoring heart rate and body temperature.



Solar Energy: Powering active cooling systems in hats.



mHealth: Providing localized health alerts via mobile.

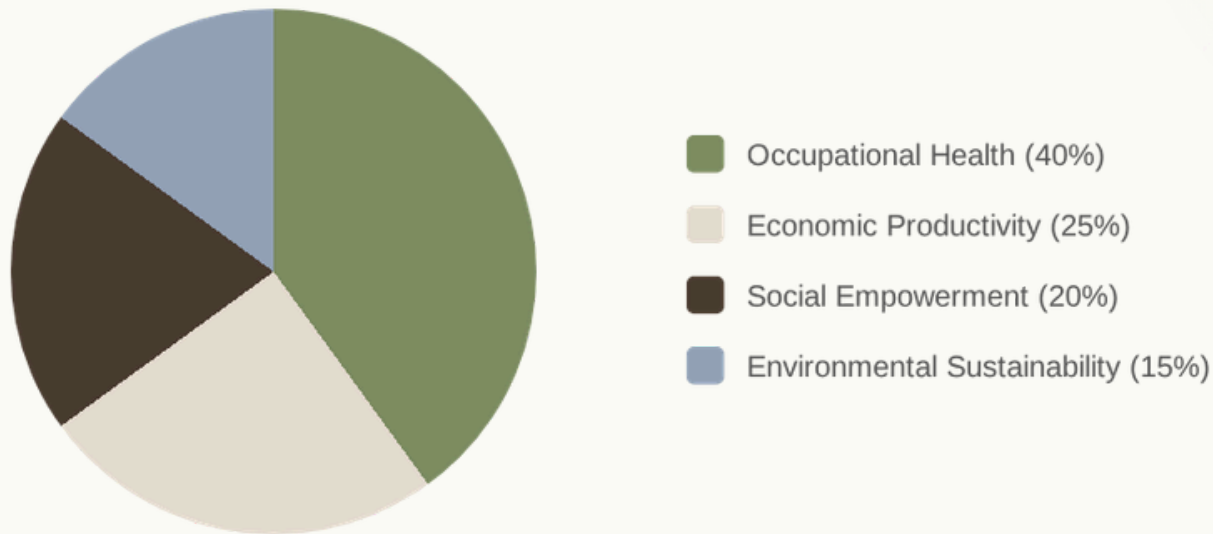


Effectiveness and Benefits Overview

Benefit Area	Specific Impact	Long-term Value
Health	Prevents heat stroke & fatigue	Reduces chronic disease risk
Economic	Improves daily productivity	Lowers family medical expenses
Social	Empowers women in decision making	Promotes gender equality
Environment	Sustainable farming practices	Responsible resource management



Impact Distribution of the Solution



The primary impact remains the immediate health protection of the woman farmer, followed by tangible economic gains.

Conclusion: A National Priority



Women farmers are the **backbone of the agricultural sector**. Their health is not just a personal matter but a requirement for national development. Our Smart Health & Safety Kit directly supports SDG 3 and aligns with the vision of the **International Year of Women Farmers 2026**. Through engineering, we can improve their quality of life fundamentally.

References

1. **United Nations Sustainable Development Goals**
<https://sdgs.un.org>
2. **World Health Organization (WHO) –**
<https://www.who.int>
3. **Food and Agriculture Organization (FAO) –**
<https://www.fao.org>
- 4 **UN Women** –<https://www.unwomen.org>
- . **Ministry of Agriculture & Farmers Welfare,**
- 5 **Government of India –**<https://agricoop.gov.in>
- 6.. **Research articles on agricultural health and safety**